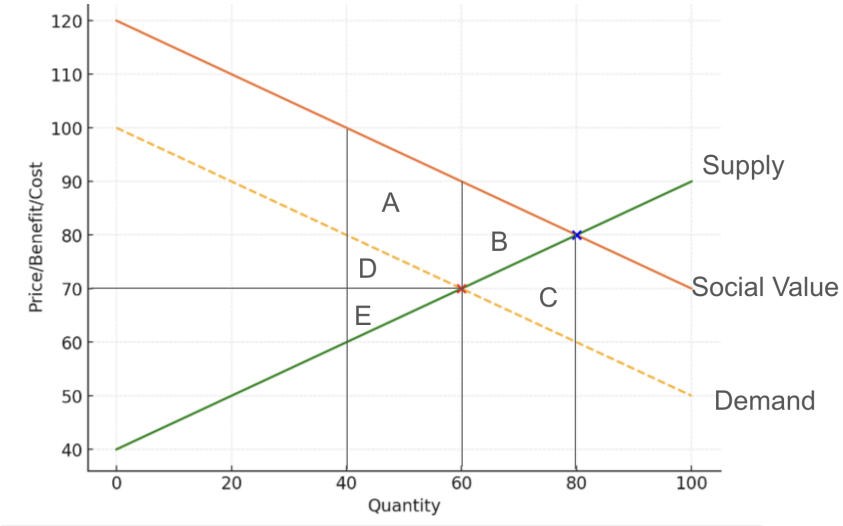
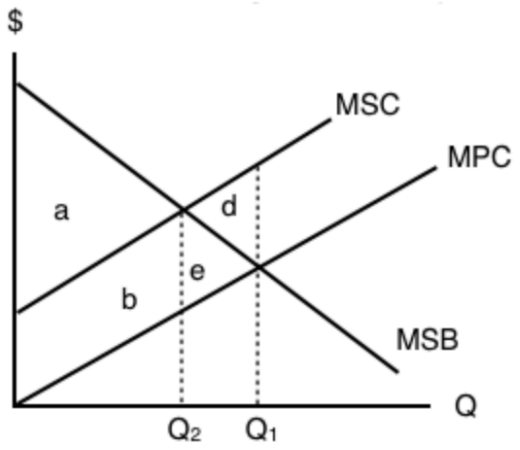
Externality Questions



1. The above graph shows a market with a positive externality. Which area represents the deadweight loss at the equilibrium under perfect competition?

a) A
b) B
c) C
d) D
e) E



2. The above graph shows a market with a negative externality, MSC is the Marginal social cost, MPC is the marginal private cost, and MSB is the Marginal social benefit which is equal to the marginal private benefit in this case since there is no positive externality. The perfectly competitive equilibrium quantity is _____ and the deadweight loss at the perfectly competitive equilibrium is _____.

a) Q_1, d
b) Q_2, e
c) Q_1, e
d) Q_2, d
e) None of the above

3. The market for good X is perfectly competitive but features positive externalities. Consumers' marginal private benefit is given by MB=210−2Q, and producers' marginal cost is MC=Q. The consumption of this good generates a marginal external benefit MEB=30. (MARK ALL THAT APPLY)

A) In perfect competition, consumers are overconsuming by 10 units.
B) In perfect competition, consumers are under-consuming by 10 units.
C) In perfect competition, consumers are overconsuming by 40 units.
D) In perfect competition, consumers are under-consuming by 40 units.
E) A $30 per unit subsidy will correct the externality.

4. For goods with a positive externality, which of the following statements is true according to economic theory (MARK ALL THAT APPLY)

a) There is deadweight loss at the market equilibrium
b) The government should provide the goods
c) Subsidizing the good could lead to the optimal quantity
d) The size of the externality is precisely estimated by policy experts
e) More of the good should be consumed than is at the market equilibrium

5. In the market for remodels, doing a remodel generates positive externalities for neighbors as it raises their property values. Therefore, in the market for remodels (**mark all that apply**),

A) the marginal social benefit curve is above and to the right of the demand curve.
B) the marginal social benefit curve is below and to the left of the demand curve.
C) the marginal social cost curve is above and to the left of the supply curve.
D) the marginal social cost curve is below and to the right of the supply curve.
E) there is no deadweight loss at the equilibrium quantity

The market for good Z is perfectly competitive but features positive externalities. Consumers' marginal private benefit is given by: $MB = 150 - 2Q$

Producers' marginal cost is: $MC = Q$

The consumption of this good generates a marginal external benefit of:

$MEB = 18$

6. Which of the following statements is true? **(MARK ALL THAT APPLY)**

A) In perfect competition, consumers are over-consuming by 6 units.
B) In perfect competition, consumers are under-consuming by 6 units.
C) In perfect competition, consumers are over-consuming by 9 units.
D) In perfect competition, consumers are under-consuming by 9 units.
E) A $18 per unit tax will correct the externality.

7. For goods with a positive externality, which of the following statements is true according to economic theory (MARK ALL THAT APPLY)

a) There is deadweight loss at the perfectly competitive market equilibrium
b) Subsidizing the good could lead to the optimal quantity
c) Taxing the good could lead to the optimal quantity
d) The size of the externality is precisely estimated by policy experts
e) More of the good should be consumed than is at the market equilibrium

8. Tradeable pollution permits are an example of **(SELECT THE BEST ANSWER)**

a) A market based solution to externalities
b) A command and control solution to externalities
c) A combination of both a market solution to externalities and a command and control one
d) Neither a command and control solution to externalities nor a market one
e) Not related to externalities

9. The market for good Q is perfectly competitive. However, it features positive externalities. Consumers' marginal private benefit is MB = 150-Q, and Producers' marginal cost is MC = Q. The consumption of this good generates a marginal external benefit MEB=20. Which of the following statements is true (mark all that apply)

a) In perfect competition, consumers are overconsuming by 10 units.
b) In perfect competition, consumers are under-consuming by 10 units
c) In perfect competition, consumers are overconsuming by 85 units.
d) In perfect competition, consumers are under-consuming by 85 units
e) A 20 dollar tax will correct the externality

10. Cap and Trade is ___

a) A market-based approach to correcting negative externalities
b) A command and control based approach to correcting negative externalities
c) Both a market and a command and control based approach to correcting negative externalities
d) Neither a market nor a command and control based approach to correcting negative externalities
e) An ancient USC tradition where students trade their graduation cap with their secret crush in the hopes of playing the finitely repeated game of marriage

11. In the market for gasoline, the use of gasoline-powered vehicles generates noise and air pollution. Therefore, in the market for gasoline (mark all that apply),

A) the marginal social benefit curve is above and to the right of the demand curve.
B) the marginal social benefit curve is below and to the left of the demand curve.
C) the marginal social cost curve is above and to the left of the supply curve.
D) the marginal social cost curve is below and to the right of the supply curve.
E) there is deadweight loss at the equilibrium quantity

The market for good **Y** is perfectly competitive but features positive externalities. Consumers' marginal private benefit is given by:

$MB=180-3Q$

Producers' marginal cost is:

$MC=Q$

The consumption of this good generates an marginal external benefit of:

$MEB=24$

12. Which of the following statements is true? **(MARK ALL THAT APPLY)**

A) In perfect competition, consumers are over-consuming by 12 units.
B) In perfect competition, consumers are under-consuming by 12 units.
C) In perfect competition, consumers are over-consuming by 6 units.
D) In perfect competition, consumers are under-consuming by 6 units.
E) A $24 per unit subsidy will correct the externality.

13. The market for gum generates negative externalities as the sound of people chewing it is annoying. Therefore, in the market for gum (**mark all that apply**)
A) the marginal social benefit curve is above and to the right of the demand curve.
B) the marginal social benefit curve is below and to the left of the demand curve.
C) the marginal social cost curve is above and to the left of the supply curve.
D) the marginal social cost curve is below and to the right of the supply curve.
E) there is no deadweight loss at the perfectly competitive equilibrium quantity

14. For goods with a negative externality, which of the following statements is true according to economic theory (MARK ALL THAT APPLY)

a) There is deadweight loss at the perfectly competitive market equilibrium
b) The government should provide the goods
c) Taxing the good could lead to the optimal quantity
d) The size of the externality is precisely estimated by policy experts
e) More of the good should be consumed than is at the market equilibrium